Department of Mathematics and Applied Mathematics

3MS Metric Spaces

Tutorial 6 **2024**

1. (a) Let d be the usual metric on $\mathbb{C}$, and let $f : (\mathbb{C}, d) \to (\mathbb{C}, d)$ be given by $f(z) = z + i$.
   i. Is f an isometry?
   ii. At which points is f continuous?

   (b) Now let d be the metric on $\mathbb{C}$ defined by

$$d(z,w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w. \end{cases}$$

   Again let $f : (\mathbb{C}, d) \to (\mathbb{C}, d)$ be given by $f(z) = z + i$.
   i. Is f an isometry?
   ii. At which points is f continuous?

2. (a) Are all isometries between metric spaces one-to-one?
   (b) Are all isometries between metric spaces onto?

3. Suppose that $f, g : (X, d_X) \to (Y, d_Y)$ are continuous functions between metric spaces. Let C be a dense subset of X. Prove that if $f(c) = g(c)$ for all $c \in C$, then $f(x) = g(x)$ for all $x \in X$. You are showing that "continuous functions agreeing on a dense subset agree on the whole space."

4. The proposition below is correct; we proved it in lectures. Explain what is wrong with this attempted simplified proof.

   Proposition Let (X, d) be a metric space and $X \times X$ have a metric which induces componentwise convergence. Then $d : X \times X \to \mathbb{R}$ is continuous.

   Attempted Proof
   Take $(x_n, y_n) \to (x, y)$.
   Then $x_n \to x$ and $y_n \to y$.
   Then $d(x_n, y_n) \leq d(x_n, x) + d(x, y) + d(y, y_n)$.
   So $\lim\limits_{n\to\infty} d(x_n, y_n) \leq \lim\limits_{n\to\infty} (d(x_n, x) + d(x, y) + d(y, y_n)) = d(x, y)$.
   Similarly, by interchanging $d(x_n, y_n)$ and $d(x, y)$ one obtains $d(x, y) \leq \lim\limits_{n\to\infty} d(x_n, y_n)$ and so $\lim\limits_{n\to\infty} d(x_n, y_n) = d(x, y)$.

5. Let d_E and d_t be the Euclidean and taxi metrics on $\mathbb{R}^2$ respectively, and define the function $f : (\mathbb{R}^2, d_E) \to (\mathbb{R}^2, d_t)$ by $f((x_1, x_2)) = (x_1, -x_2)$, for $(x_1, x_2) \in \mathbb{R}^2$.

   (a) Is f an isometry?

   (b) Is f continuous?

6. Let d_E denote the usual metric on $\mathbb{R}$ or $\mathbb{R}^2$.

   (a) Show that there are exactly two isometries $f : (\mathbb{R}, d_E) \to (\mathbb{R}, d_E)$ which leave a given element $b \in \mathbb{R}$ fixed.

   (b) Does the same result hold for isometries $f : (\mathbb{R}^2, d_E) \to (\mathbb{R}^2, d_E)$?

The next two questions concern sets and set functions only; no metrics or continuity. Let $f : X \to Y$ be a function from the set X to the set Y. Let $A, A_i \subseteq X$ for $i \in I$ and $B, B_i \subseteq Y$ for all $i \in I$.

Recall that $f(A) = \{f(a) : a \in A\}$ and $f^{-1}(B) = \{x \in X : f(x) \in B\}$.

7. **Properties of inverse images**
   Prove a selection of the following; enough so that you're comfortable you know how this works. You'll see that inverse images are very well-behaved.

   (a) $f^{-1}(\bigcap\limits_{i \in I} B_i) = \bigcap\limits_{i \in I} f^{-1}(B_i)$

   (b) $f^{-1}(\bigcup\limits_{i \in I} B_i) = \bigcup\limits_{i \in I} f^{-1}(B_i)$

   (c) $f^{-1}(\emptyset) = \emptyset$

   (d) $f^{-1}(Y) = X$

   (e) $f^{-1}(B') = (f^{-1}(B))'$

8. **Properties of direct images**
   Direct images are not so well-behaved. Again, prove a selection of these; do at least one requiring a counter-example.

   (a) $f(\bigcap\limits_{i \in I} A_i) \subseteq \bigcap\limits_{i \in I} f(A_i)$ but equality need not hold.

   (b) $f(\bigcup\limits_{i \in I} A_i) = \bigcup\limits_{i \in I} f(A_i)$

   (c) $f(\emptyset) = \emptyset$

   (d) $f(X) \subseteq Y$ but equality need not hold.

   (e) There is no automatic relation between $f(A')$ and $(f(A))'$.